Consider a population of some kind. For concreteness, let's say it is a population of rabbits. The number of individuals in the population, in general, will change over time.

1.) List at least 3 ways in which the number of rabbits in the population might change over time.

Our goal is to find a mathematical model which describes how the population changes over time. We begin by identifying the independent and dependent variables:

$t =$ time (the independent variable)

$P =$ the number of individuals in the population (the dependent variable)

In terms of these variables, our goal is to find a function which gives $P(t)$ for each time t. In order to do so, we need to make some assumptions about how population evolves over time based on our knowledge of the population as determined through data, experiments, etc.

2.) A physically reasonable assumption for many populations is that the rate of population growth is proportional to the size of the population.

(a) Explain why this assumption makes sense for a population of rabbits.

(b) Name some other populations for which this assumption makes sense.

(c) Write this assumption as a mathematical equation in terms of the variables identified above.

The rate of population growth is the derivative $\frac{dP}{dt}$. Our assumption that the rate of growth of the population is proportional to the population size can be written as

$$\frac{dP}{dt} = kP \tag{1}$$

where k is the proportionality constant. This is an example of a *differential equation*, which is an equation which relates an unknown function to one or more of its derivatives. Equation (1) is said to be a *first-order* differential equation, because the order of the highest-derivative appearing in the equation is one.

3.) Having formulated a model, let's look at its consequences. If we assume that the population is never 0, then $P(t) > 0$ for all t. If $k > 0$, what does Equation (1) imply under this assumption?

4.) Let's now try to find a function P which satisfies Equation (1). Can you think of a function whose derivative is a constant multiple of itself?

5.) Show that the function Ce^{kt} satisfies Equation (1). We will prove later that this is the only solution.

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We have just seen that any exponential function of the form $P(t) = Ce^{kt}$, where C is any real number, is a solution of Equation (1). We therefore find a *family* of solutions to Equation (1) whose graphs are shown to the left in the figure below. Since $P(t) > 0$ for all t, we are only interested in the solutions with $C > 0$. Also, we are only concerned with values of t greater than the initial time $t = 0$. The graphs of the physically meaningful solutions are shown to the right in the figure below.

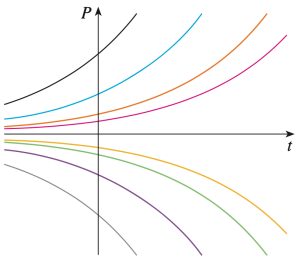


FIGURE 1
The family of solutions of $dP/dt = kP$

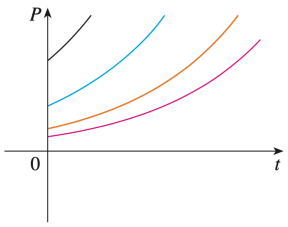


FIGURE 2
The family of solutions $P(t) = Ce^{kt}$ with $C > 0$ and $t \geqslant 0$

6.) Consider $P(0)$. What is the physical interpretation of C?

In applications of differential equations, one is usually not as interested in the entire family of solutions (the *general solution*) as one is in a particular solution specified by $P(0) = P_0$. This is called an *initial condition*, and the problem of solving a differential equation subject to an initial condition is called an *initial-value problem*. Geometrically, when we impose an initial condition, we look at the family of solution curves and pick the one which passes through the point $(0, P_0)$. Physically, this corresponds to measuring the state of the system at time 0 and using the solution of the initial-value problem to predict the future behavior of the system.

7.) How accurately do you expect our model to describe the population? Have we made any unrealistic assumptions and/or neglected any important features of the population? Discuss some limitations of our model.

Equation (1) is appropriate for modeling population growth under ideal conditions. A more realistic model must reflect the fact that a given environment has limited resources. Many populations start by increasing in an exponential manner, but the population levels off when it approaches its *carrying capacity* (or decreases toward M if it ever exceeds M). For a model to take into account both trends, we make two assumptions:

(i) $\frac{dP}{dt} \approx kP$ if P is small compared to M (Initially, the growth rate is proportional to P.)

(ii) $\frac{dP}{dt} < 0$ if $P > M$ (P decreases if it ever exceeds M.)

8.) Show that the equation

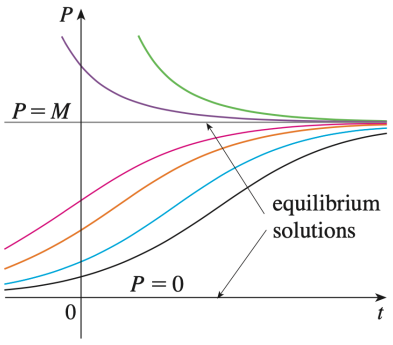
$$\frac{dP}{dt} = kP \cdot \left(1 - \frac{P}{M}\right) \tag{2}$$

satisfies both assumptions above.

Equation (2) is called the *logistic differential equation* and was proposed by the Dutch mathematical biologist Pierre-Francois Verhulst in the 1840s as a model for the world population growth. We will soon develop techniques that enable us to find explicit solutions of the logistic equation, but for now we can deduce qualitative characteristics of the solutions directly from Equation (2).

9.) Show that the constant functions $P(t) = 0$ and $P(t) = M$ satisfy Equation (2). Interpret these solutions physically. These two solutions are called *equilibrium solutions*.

10.) The graphs of the solutions to the logistic equation are shown below. Explain why these are consistent with Equation (2). (E.g., what happens if $0 < P(0) < M$? What happens if $P(0) > M$?)

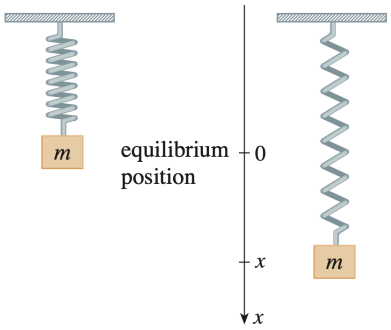


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Let's now look at an example of a mathematical model from physics. Consider the motion of an object of mass $m > 0$ at the end of a spring, the other end of which is attached to a wall. The force on the mass due to the stretched (or compressed) spring is given by *Hooke's law*, which says that

$$F = -kx$$

where x is the displacement of the spring from its equilibrium position, and k is a positive constant (called the *spring constant*) which describes the stiffness of the spring. The negative sign is due to the fact the force is directed in the opposite direction relative to which the string is stretched (or compressed).



In the absence of air resistance or friction, the displacement x of the mass from the equilibrium position of the spring as a function of time satisfies *Newton's Second Law* (Force = mass $\times$ acceleration), which can be written as

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x \tag{3}$$

11.) What is the order of the differential equation (3)?

12.) Can you think of a function whose second derivative is proportional to itself, but with opposite sign?

13.) Let $\omega = \sqrt{\frac{k}{m}}$, so that Equation (3) becomes

$$\frac{d^2x}{dt^2} = -\omega^2 x. \tag{4}$$

(a) Show that $x(t) = c_1\cos(\omega t) + c_2\sin(\omega t)$ satisfies Equation (4).

(b) Solve Equation (4) subject to the *two* initial conditions $x(0) = A$ and $\frac{dx}{dt}(0) = 0$.

(c) Sketch the graph of your solution in part (b) when $A > 0$.

(d) What is the physical interpretation of A and ω?